Nguyen Vu Binh Duong

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Education

University of Engineering and Technology, Vietnam National University (VNU-UET) — Hanoi, Vietnam

M.Sc. in Software Engineering — *Dec 2023 – June 2025*

- **GPA**: 3.87/4.00
- **Master's Thesis**: AI-based Software System for Early Diagnosis and Detection of Alzheimer's Disease in Vietnamese People

B.Sc. in Computer Science — *Sep 2019 – June 2023*

- **GPA**: 3.60/4.00
- **Bachelor's Thesis**: Improving the Performance of Unit Test Case Execution for C/C++ Projects

Research Experience

International University, Vietnam National University Ho Chi Minh City — HCMC, Vietnam

Brain Health Lab, School of Biomedical Engineering

Researcher — *Jan 2024 – Present*

Alzheimer's Biomarker Cross-border Development (ABCD) Project, Vietnamese-Swiss Joint Research Programme
Supervisor: Assoc. Prof. Huong Ha

- Built an ML framework (predictor + explainer modules) for AD classification across **113 Vietnamese subjects**, achieving **81.4% accuracy** with XGBoost and **78.8% accuracy / 87.2% AUC** with the Generalized Additive Model.
- Compared brain atrophy structural features across **2 cohorts** (Vietnamese N=76 vs. ADNI N=80), revealing a **10–25% accuracy drop** in cross-dataset validation, underscoring the need for population-specific models.

Project Unveiling Heterogeneity in Early Alzheimer's Disease
Supervisor: Assoc. Prof. Huong Ha

- Preprocessed MRI scans with FreeSurfer (104 volumetric features) and applied CIMLR clustering to identify **5 distinct subgroups** within 648 MCI/CN subjects.
- Characterized clusters via statistical analysis of **demographics, cognitive scores, and brain volumes**, revealing **two mild-atrophy** clusters (enriched with CN) and **one severe-atrophy** cluster (enriched with MCI).

University of Engineering and Technology, Vietnam National University — Hanoi, Vietnam

Department of Software Engineering, Faculty of Information Technology

Lab Instructor/Research Assistant — *Sep 2023 – Present*

Undergraduate Research Assistant — *Feb 2021 – June 2023*

AKAUTAUTO: a proprietary software used for C/C++ unit testing with automated unit test generation. Supervisor: Assoc. Prof. Pham Ngoc Hung

- Directed the research group of **15 members** and facilitated weekly technical reviews with the joint development team at **FPT Software**.
- Provided training and technical consultation to group members.
- Implemented a pairwise test case generation algorithm and integrated it with the classification tree feature.
- Developed a stubbing method for **built-in and external** dependencies, enabling unit testing of **80% previously untestable** modules.

Publications & Presentations

Vu, D.-T., **Duong, N. V. B.**, Chén, O. Y., Ha, H., et al. Bridging Kernel and Feature Spaces: A New Approach to Accurate and Interpretable Alzheimer's Disease Assessment. *(In preparation for submission to Journal of Biomedical Informatics)*

Duong, N. V. B., Vu, D.-T., Can, D.-C., Nguyen, H. D., Nguyen, B. T., Chén, O. Y., & Ha, H. (2025). A Comparison of Machine Learning Methods for Alzheimer's Disease Classification in Vietnamese Patients. Presented at the 14th International Symposium on Information and Communication Technology (SOICT 2025), Vietnam. *(To appear in Communications in Computer and Information Science)*

Duong, N. V. B., Ha, H., Ngo, H. P., Vu, D. T., Can, D.-C., Nguyen, H. D., Bui, L. C., Nguyen, B. T., & Chén, O. Y. (2025). Machine Learning-Based Analysis of Brain Atrophy Patterns in Alzheimer's Disease: A Comparative Study Between ADNI and Vietnamese Cohorts. Poster presented at the International Conference on Korean Dementia Association & Asia Summit Against Dementia (IC-KDA & ASAD 2025), Korea.

Nguyen, M., **Duong, N. V. B.**, Duc, V. T., Phan, C., Phan, T., & Ha, H. (2025). Unveiling Heterogeneity in Early Alzheimer's Disease: A Data-Driven Exploration of Mild Cognitive Impairment and Normal Cognition. *IFMBE Proceedings*, 408–427. https://doi.org/10.1007/978-3-031-90194-2_29

Lam Nguyen Tung, **Duong, N. V. B.**, Khoi Nguyen Le, & Pham Ngoc Hung. (2024). Automated test data generation and stubbing method for C/C++ embedded projects. *Automated Software Engineering*, 31(2). https://doi.org/10.1007/s10515-024-00449-6

Awards and Honors

IC-KDA Asian Society Against Dementia Conference Travel Grant, Korea	May 2025
Vingroup Innovation Foundation Master's Scholarship, Vietnam	Dec 2023 – Jan 2026
Second prize, The Scientific Research Conference for Undergraduates at VNU-UET, Vietnam	May 2023

Technical Skills

Neuroimaging & Signal Processing: CAT12, FreeSurfer, CONN
Machine Learning: XGBoost, GAM, scikit-learn
Software Development: Python, JavaScript, C++, Automated Software Testing
Research Methods: Cohort Studies, Cross-dataset Validation
Tools: Git, Linux, Docker, Jira, LaTeX, Microsoft Office.
Languages: Vietnamese (native), English (IELTS 7.5 and TOEFL IBT 106/120), Chinese (beginner)

References

Assoc. Prof. Pham Ngoc Hung
Vice Dean | Faculty of Information Technology, University of Engineering and Technology, Vietnam National University
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Assoc. Prof. Huong Ha
Chair | Department of Tissue Engineering and Regenerative Medicine, School of Biomedical Engineering, International University, Vietnam National University Ho Chi Minh City
Email: htthuong@hcmiu.edu.vn